Week #11: [Cranial Nerve Anatomy]

Cranial Nerve Anatomy & Function

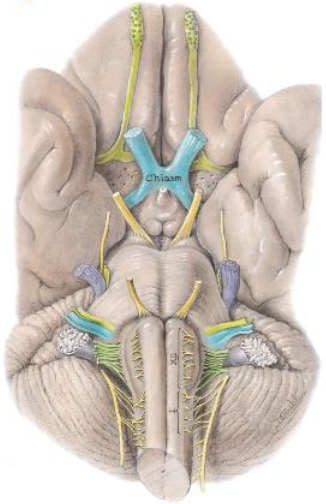
- Describe cranial nerve anatomy and function
- Predict symptoms of cranial nerve pathology

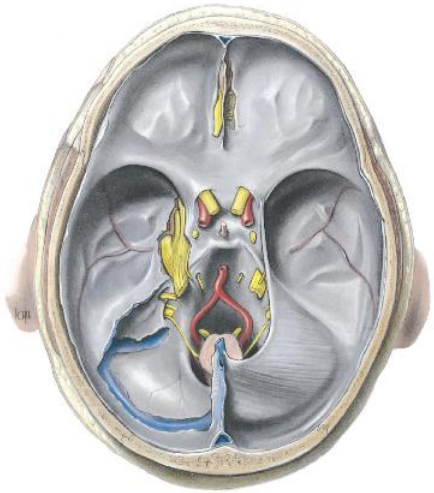
Cranial Nerve Anatomy & Function

- Describe cranial nerve anatomy and function
- Predict symptoms of cranial nerve pathology

What are the cranial nerves?

- Bundles of nerve fibers that exit the cranial cavity
- 12 pairs in number





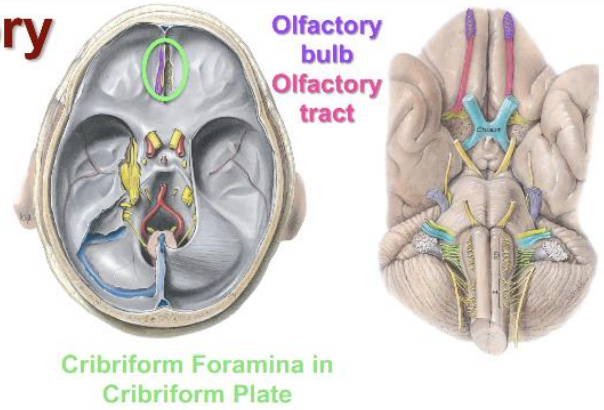
The Cranial Nerves

Sensory
Motor
Both

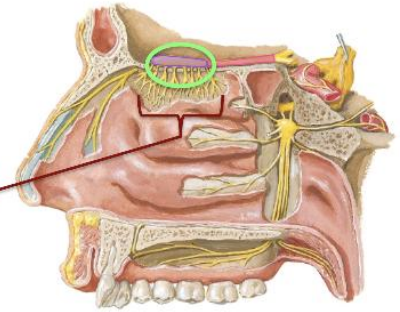
CN	Name	Mnemonic	Fibers
CN I	Olfactory nerve	Oh	Some
CN II	Optic nerve	Oh	Say
CN III	Oculomotor nerve	Oh	Marry
CN IV	Trochlear nerve	To	Money
CN V	Trigeminal nerve	Touch	But
CN VI	Abducent nerve	And	My
CN VII	Facial nerve	Feel	Brother
CN VIII	Vestibulocochlear nerve	Very	Says
CN IX	Glossopharyngeal nerve	Good	Big
CN X	Vagus nerve	Velvet	Brains
CN XI	Spinal Accessory nerve	AH	Matter
CN XII	Hypoglossal nerve	AH	More

CN I: Olfactory Nerves

- **Fibers:**
  - Special Sensory
- **General Function:**
  - Olfaction (smell)
- **Test:**
  - Smell







B. Intracranial Path of the Cranial Nerves:

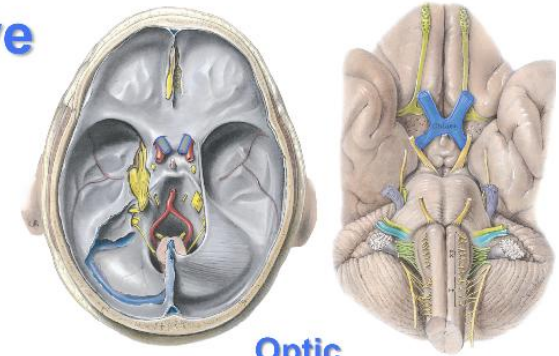
1. Olfactory nerves (CN I)

a. Axons from the olfactory areas of the nasal cavities traverse cribriform foramina of the ***cribriform plate*** (ethmoid bone) to synapse on neurons in the *olfactory bulbs* of the brain. Axons of these neurons form *olfactory tracts* that terminate at the cerebral cortex.

b. ***Special Sensory:*** Smell

CN II: Optic Nerve

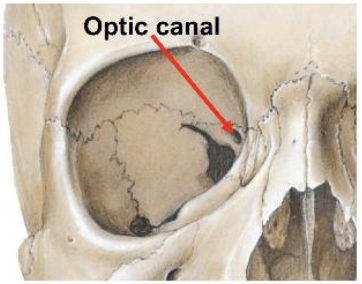
- **Fibers:**
  - Special Sensory
- **General Function:**
  - Vision
- **Test:**
  - Visual Acuity
  - Visual Field
  - Pupillary Light Reflex
  - Ophthalmoscope











2. Optic nerve (CN II)

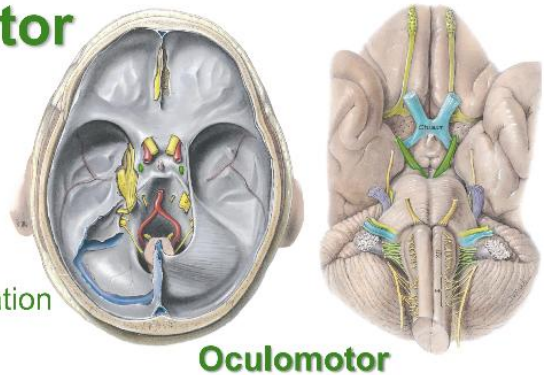
a. Extends posteriorly from eyeball to exit through the **optic canal** & enter the middle cranial fossae.

b. Fibers decussate (cross) in the **optic chiasm** & join uncrossed fibers to form the ***optic tract***. Fibers in the optic tract terminate in the lateral geniculate bodies of the thalamus. Fibers are eventually relayed to the visual cortices of the occipital lobes of the brain.

c. ***Special Sensory:*** Vision

CN III: Occulomotor Nerve

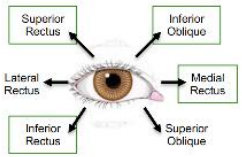
- **Fibers:**
  - Motor
  - Parasympathetic
- **General Function:**
  - Muscles of Eye
  - Constriction & Accommodation
- **Test:**
  - Eye Movements
  - Pupillary Light Reflex

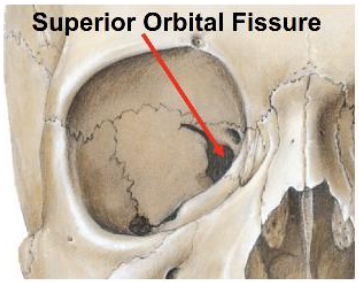


Oculomotor Nerves









3. Oculomotor nerve (CN III)

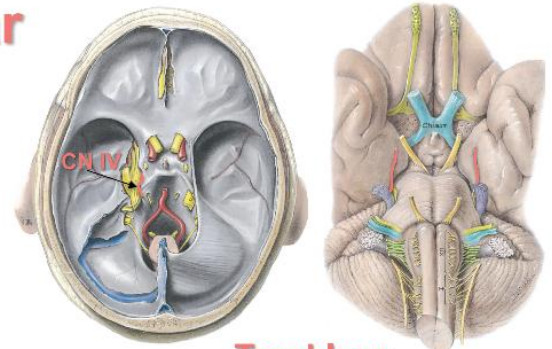
a. Originates from the midbrain, pierces the dura lateral to the diaphragma sellae, to run through the lateral wall of the cavernous sinus. Enters the orbit via the **superior orbital fissure.**

b. ***Motor:***
   i. levator palpebrae superioris
   ii. superior rectus, medial rectus
   iii. inferior rectus
   iv. inferior oblique muscles

c. ***Parasympathetics:*** **sphincter pupillae muscle** (constricts pupil) and the ciliary muscle/body (allows lens to thicken for near vision = **accommodation**).

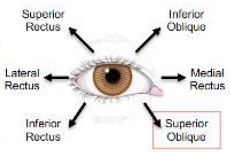
CN IV: Trochlear Nerve

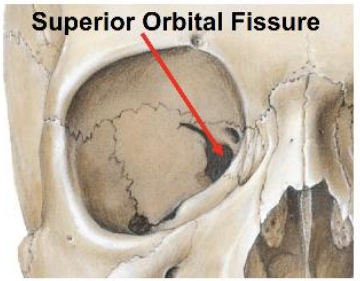
- **Fibers:**
  - Motor
- **General Function:**
  - One Muscle of Eye
- **Test:**
  - Eye Movement



Trochlear Nerves







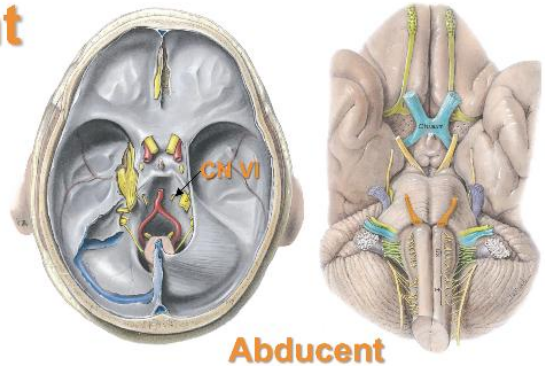
4. Trochlear nerve (CN IV)

a. Runs the longest intracranial course as it originates from the **posterior** side of the midbrain. Pierces dura near the posterior clinoid process & runs along the lateral wall of the cavernous sinus. Enters the orbit through the **superior orbital fissure.**

b. ***Motor:*** superior oblique muscle

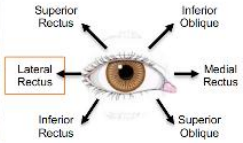
CN VI: Abducent Nerve

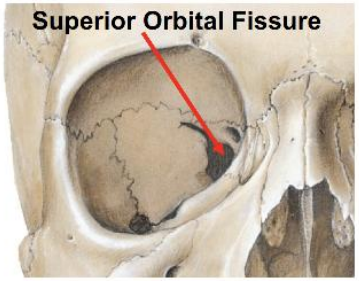
- **Fibers:**
  - Motor
- **General Function:**
  - One Muscle of Eye
- **Test:**
  - Eye Movement



Abducent Nerves







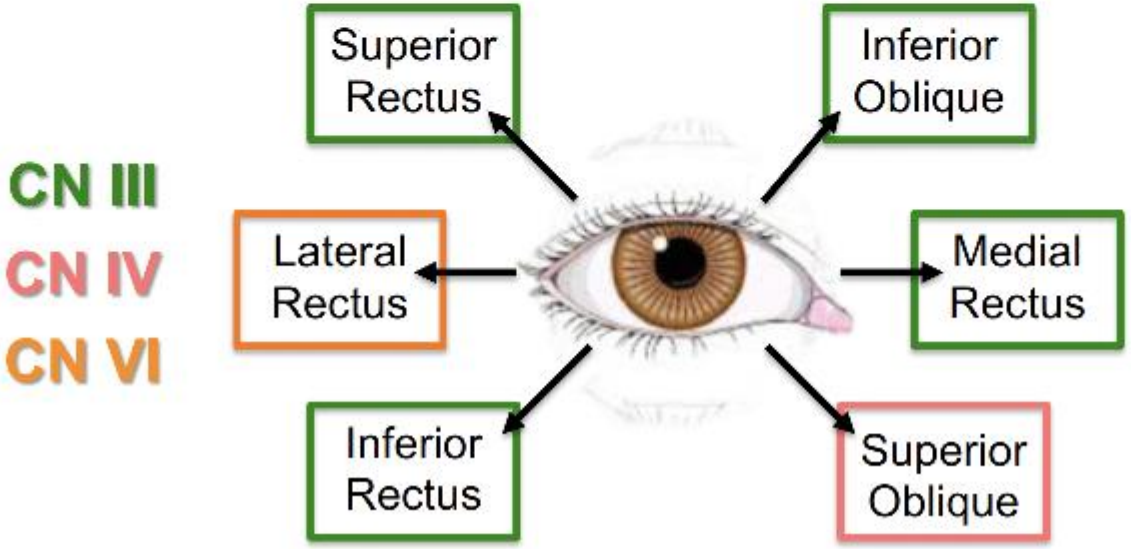
6. Abducent nerve (CN VI)

a. Emerges from the brainstem between the pons & medulla. Runs longest *intradural* course of the cranial nerves, piercing the dura on the clivus, then coursing through the middle of the cavernous sinus. Exits cranial cavity through the **superior orbital fissure.**

b. ***Motor:*** lateral rectus

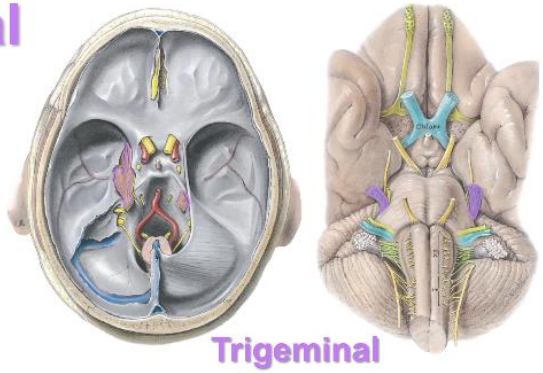
Testing Extra-ocular Eye Movements



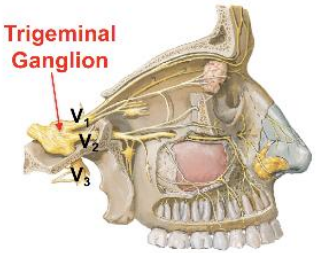


CN V: Trigeminal Nerve

- **Three Divisions:**
  - V_1: Ophthalmic Nerve
  - V_2: Maxillary Nerve
  - V_3: Mandibular Nerve
- **Fibers:**
  - Both
    - V_1 – Sensory
      - Sympathetic
    - V_2 – Sensory
    - V_3 - Sensory & Motor



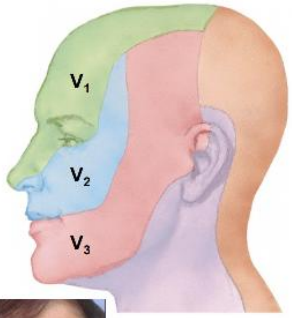
Trigeminal Nerves



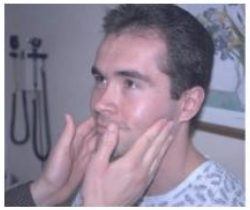
CN V: Trigeminal Nerve

- **Three Divisions:**
  - V_1: Ophthalmic Nerve
  - V_2: Maxillary Nerve
  - V_3: Mandibular Nerve
- **General Function:**
  - V_1 – Sensory to upper face & cornea, & sympathetic to pupil dilators
  - V_2 – Sensory to mid face
  - V_3 - Sensory to lower face, motor to muscles of mastication
- **Test:**
  - V_1, V_2, V_3 – Touch in 3 sensory locations
  - V_1 – Corneal Reflex
  - V_3 – Muscles of Mastication









5. Trigeminal nerve (CN V)

a. Emerges from the pons by a large sensory root & small motor root.
   i. Sensory root forms the **trigeminal ganglion** (where its cell bodies are located) under the dura.
   ii. Motor root of CN V runs parallel to the sensory root, bypasses the ganglion, & becomes part of the ***mandibular nerve*** (CN V_3).

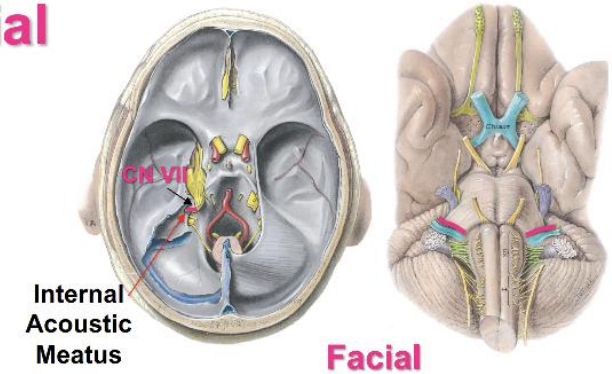
b. Trigeminal ganglion forms 3 nerves/divisions:
   i. Ophthalmic nerve (CN V_1)
      1. Exits cranial cavity via ***superior orbital fissure***
      2. ***Sensory:*** skin of forehead, scalp, eyelids, nose; cornea; paranasal sinus & superior nasal mucosa

   ii. Maxillary nerve (CN V_2)
      1. Exits cranial cavity via ***foramen rotundum***
      2. ***Sensory:*** skin of the upper jaw, including the upper lip, maxillary teeth, inferior nasal mucosa, maxillary sinus, and palate

   iii. Mandibular nerve (CN V_3)
      1. Exits cranial cavity via ***foramen ovale***
      2. ***Sensory:*** skin over the mandible, including the lower lip, buccal, parotid, & temporal regions of face, external ear, mandibular teeth, temporomandibular joint, mucosa of mouth, and anterior 2/3 of tongue (touch & temperature)
      3. ***Motor:*** muscles of mastication, mylohyoid, anterior belly of the digastric, & tensor veli palatini

CN VII: Facial Nerve

- **Fibers:**
  - Both
  - Autonomic
- **General Function:**
  - Motor to muscles of facial expression
  - Special sensory – taste to ant. 2/3rd tongue
  - Parasympathetic to the lacrimal, nasal, palatine, and salivary glands
  - Sympathetic to the lacrimal, nasal, and palatine glands
- **Test:**
  - Facial Expressions
  - Taste
  - Corneal Reflex



Facial Nerves

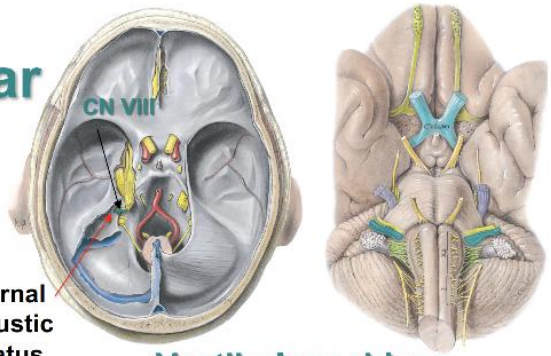


7. Facial nerve (CN VII)

a. Arises from brainstem between the pons & medulla and exits cranial cavity via the **internal acoustic meatus**. It runs through the ***facial canal*** in the petrous part of the temporal bone and exits the cranium via the ***stylomastoid foramen*** (longest intraosseous course).
b. ***Motor:*** muscles of facial expression, stylohyoid, posterior belly of digastric, stapedius
c. ***Special Sensory:*** Taste sensation from the **anterior 2/3 of tongue**
d. ***Sensory:*** skin in and around the auricle (outer ear)
e. ***Parasympathetics:*** to the **lacrimal gland** (tears) and **sublingual** & **submandibular** glands

CN VIII: Vestibulocochlear Nerve

- **Fibers:**
  - Sensory
- **General Function:**
  - Hearing
  - Balance
- **Test:**
  - Hearing



Vestibulocochlear Nerves

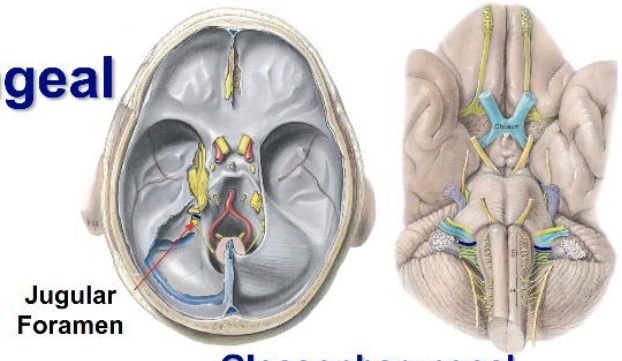


8. Vestibulocochlear nerve (CN VIII)

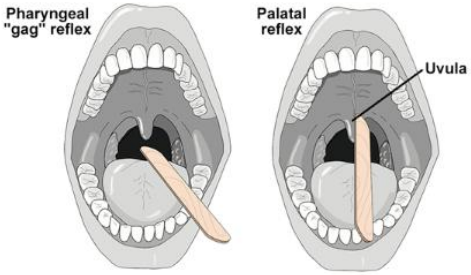
a. Originates from the brainstem between the pons & medulla and enters the **internal acoustic meatus**, where it divides into the vestibular (equilibrium & motion) and cochlear nerves (hearing).
   i. CN VIII does not exit the cranium.
b. ***Special Sensory:*** hearing, equilibrium, & motion

CN IX: Glossopharyngeal Nerve

- **Fibers:**
  - Both
  - Parasympathetic
- **General Function:**
  - Motor to one muscle of the pharynx
  - General sensory to tongue & palate
  - Special sensory – taste to post. 1/3rd tongue
  - Parasympathetic to parotid gland
- **Test:**
  - Gag Reflex



Glossopharyngeal Nerves

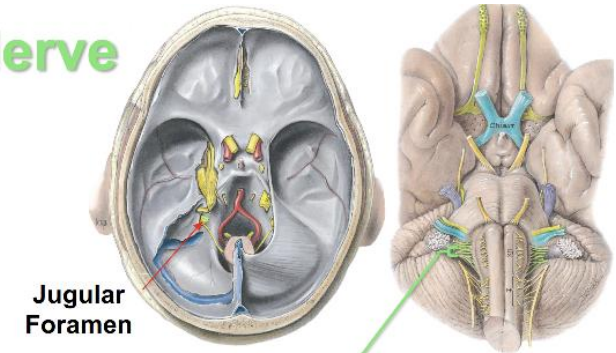


9. Glossopharyngeal nerve (CN IX)

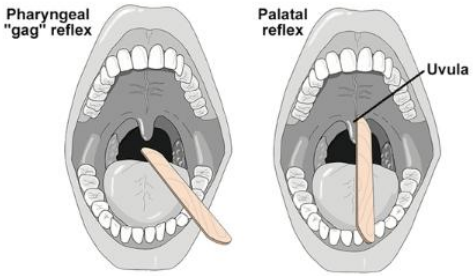
a. Emerges from lateral aspect of medulla and exits cranial cavity through jugular foramen.
b. ***Motor:*** stylopharyngeus muscle
c. ***Sensory:*** to pharynx (touch, pain, and temperature sensations); sensation from the carotid sinus (changes in blood pressure) and carotid body (changes in blood gas levels – O_2 & CO_2)
d. ***Special sensory:*** Taste to the posterior 1/3 of tongue
e. ***Parasympathetics:*** parotid gland secretion

CN X: Vagus Nerve

- **Fibers:**
  - Both
  - Parasympathetic
- **General Function:**
  - Motor to palate and laryngeal muscles
  - General sensory to pharynx & larynx
  - Parasympathetic to thoracic and abdominal viscera
- **Test:**
  - Saying "Ahhh"
  - Gag Reflex
  - Phonation



Vagus Nerve

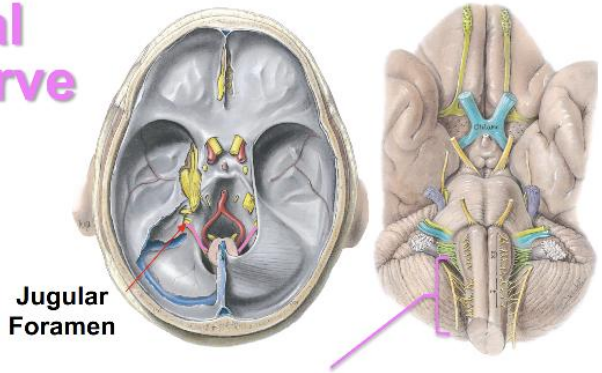


10. Vagus nerve (CN X)

a. Arises from the medulla and leaves the cranial cavity through the **jugular foramen.**
b. Runs the longest course with the most extensive distribution of all the cranial nerves.
c. ***Motor:*** pharyngeal, palatine, most intrinsic larynx muscles
d. ***Parasympathetic:*** thoracic & abdominal viscera to the left colic flexure
e. ***General sensory:*** to the inferior pharynx & larynx
f. ***Special sensory:*** Taste from root of the tongue

CN XI: Spinal Accessory Nerve

- **Fibers:**
  - Motor
- **General Function:**
  - Motor to trapezius & sternocleidomastoid
- **Test:**
  - Function of trapezius & sternocleidomastoid





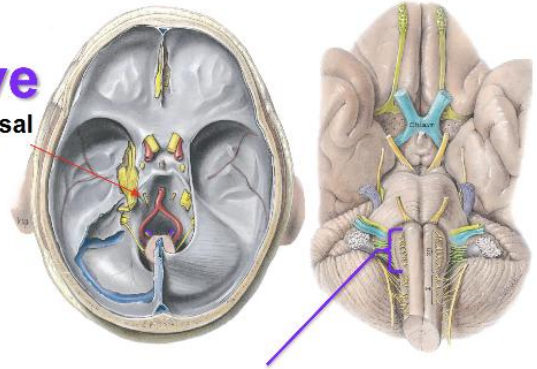
11. Spinal accessory nerve (CN XI)

a. Arises from the **superior 5-6 cervical segments of the spinal cord**, ascends to enter the cranial cavity via the **foramen magnum**, and then exits via the **jugular foramen.**

b. ***Motor:*** trapezius & sternocleidomastoid

CN XII: Hypoglossal Nerve

- **Fibers:**
  - Motor
- **General Function:**
  - Motor to the tongue
- **Test:**
  - Function of the tongue





12. Hypoglossal nerve (CN XII)

a. Arises from medulla and leaves the cranial cavity through the **hypoglossal canal.**

b. ***Motor:*** intrinsic & extrinsic tongue muscles

Summary

CN	Name	General Function
CN I	Olfactory nerve	**Smell**
CN II	Optic nerve	**Vision**
CN III	Oculomotor nerve	**Eye Movement**
CN IV	Trochlear nerve	**Eye Movement**
CN V	Trigeminal nerve	**Sensory to Face**
CN VI	Abducent nerve	**Eye Movement**
CN VII	Facial nerve	**Facial Expression**
CN VIII	Vestibulocochlear nerve	**Hearing**
CN IX	Glossopharyngeal nerve	**Gag Reflex**
CN X	Vagus nerve	**“Ahhhhhh”**
CN XI	Spinal Accessory nerve	**Traps & SCM**
CN XII	Hypoglossal nerve	**Tongue Movement**